Electrolysis

Answer key with worked solutions

A. Understanding

1. Concepts.

> (a) Using electricity to drive a **non-spontaneous** redox reaction. (b) An electrolytic cell **uses** electrical energy, a galvanic cell **produces** it; in electrolysis the anode is **positive**, in a galvanic cell it is negative. (c) The **ions must be free to move** to the electrodes; in a solid ionic compound they are fixed in the lattice.

B. Predicting products

2. Molten compounds.

> (a) Cathode **potassium**, anode **bromine** (b) cathode **magnesium**, anode **chlorine** (c) cathode **aluminium**, anode **oxygen**

3. Aqueous solutions.

> (a) Cathode **copper**, anode **chlorine** (b) cathode **hydrogen**, anode **oxygen** (c) cathode **silver**, anode **oxygen** (d) cathode **hydrogen**, anode **iodine**

4. Half-equations.

> (b) Cathode: $2\,H_2O + 2\,e^- \longrightarrow H_2 + 2\,OH^-$; anode: $2\,H_2O \longrightarrow O_2 + 4\,H^+ + 4\,e^-$. (c) Cathode: $Ag^+ + e^- \longrightarrow Ag$; anode: $2\,H_2O \longrightarrow O_2 + 4\,H^+ + 4\,e^-$.

5. Copper refining.

> The **impure copper anode** is oxidized and dissolves: $Cu \longrightarrow Cu^{2+} + 2\,e^-$. At the **pure copper cathode**, $Cu^{2+} + 2\,e^- \longrightarrow Cu$ deposits pure copper. Less reactive impurities (silver, gold) do not dissolve and fall to the bottom as **anode sludge**; more reactive ones dissolve but stay in solution.

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C. Faraday's law

6. Silver plating.

> $Q = (0.500\ \text{C/s})(30.0\ \text{min} \times 60\ \text{s/min}) = 900.\ \text{C}$; $n(e^-) = \dfrac{900.\ \text{C}}{96\,485\ \text{C/mol}} = 9.328 \times 10^{-3}\ \text{mol}$
>
> 1 mol e⁻ per mol Ag: $m = 9.328 \times 10^{-3}\ \text{mol} \times 107.87\ \text{g/mol} = \mathbf{1.01\ g}$

7. Copper deposition time.

> $n(\text{Cu}) = \dfrac{2.00\ \text{g}}{63.55\ \text{g/mol}} = 0.03147\ \text{mol}$; $n(e^-) = 2 \times 0.03147\ \text{mol} = 0.06294\ \text{mol}$
>
> $Q = 0.06294\ \text{mol} \times 96\,485\ \text{C/mol} = 6073\ \text{C}$; $t = \dfrac{6073\ \text{C}}{1.50\ \text{C/s}} = 4049\ \text{s} = \mathbf{67.5\ min}$

8. Electrolysis of water.

> $Q = (0.500\ \text{C/s})(7200\ \text{s}) = 3600\ \text{C}$; $n(e^-) = \dfrac{3600\ \text{C}}{96\,485\ \text{C/mol}} = 0.03731\ \text{mol}$
>
> H_2 (2 e⁻ each): $n = 0.01866\ \text{mol}$; $V = \dfrac{(0.01866\ \text{mol})(0.08206\ \text{L atm mol}^{-1}\,\text{K}^{-1})(298.15\ \text{K})}{1.00\ \text{atm}} = \mathbf{0.456\ L}$
>
> O_2 (4 e⁻ each): $n = 9.328 \times 10^{-3}\ \text{mol}$; $V = \mathbf{0.228\ L}$ (half the volume of hydrogen, as expected from $2\,H_2O \longrightarrow 2\,H_2 + O_2$)

D. Challenge

9. Energy for aluminium.

> (a) $n(\text{Al}) = \dfrac{1.00 \times 10^3\ \text{g}}{26.98\ \text{g/mol}} = 37.06\ \text{mol}$; $n(e^-) = 3 \times 37.06\ \text{mol} = 111.2\ \text{mol}$
>
> $Q = 111.2\ \text{mol} \times 96\,485\ \text{C/mol} = \mathbf{1.07 \times 10^7\ C}$
>
> (b) $E = QV = (1.073 \times 10^7\ \text{C})(4.5\ \text{V}) = \mathbf{4.8 \times 10^7\ J}$; $\dfrac{4.83 \times 10^7\ \text{J}}{3.6 \times 10^6\ \text{J/kWh}} = \mathbf{13\ kWh}$ per kilogram
>
> (c) Aluminium production needs **huge amounts of electricity**, which is the main cost, so smelters are built where electricity is cheap and plentiful (and, with hydroelectric power, low in carbon emissions).

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